

CSC3066

Deep Learning

Multilayer Perceptron

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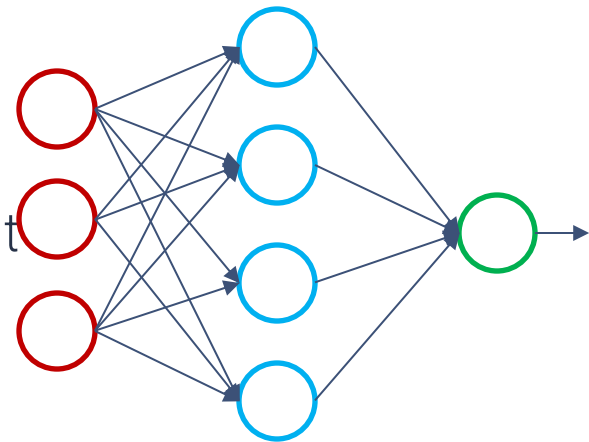
What's to come today

- ❑ Multilayer Perceptron (MLP)



Artificial Neural Network: Multilayer Perceptron

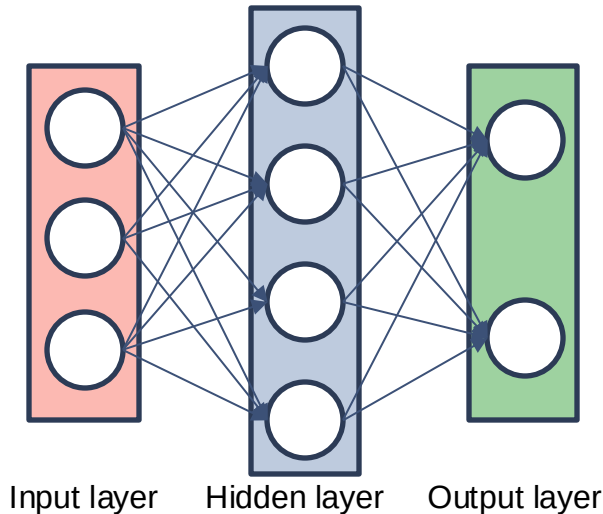
- ❑ We can improve things by increasing the number of neurons and layers in the network - Multilayer Perceptron (MLP)
- ❑ MLP: consists of at least 3 layers of nodes:
 - ❑ **Input layer:** input values
 - ❑ **Hidden layer(s):** a layer that is neither the input layer nor the output layer
 - ❑ **Output layer:** neurons from which we read the results





Multilayer Perceptron: Fundamental Concepts

- ❑ Input units: receive various form of information from the outside world that the network will attempt to learn about.
- ❑ Hidden units: perform mathematical calculations on the input values
- ❑ Output units: provide response of the network to the information provided as input.





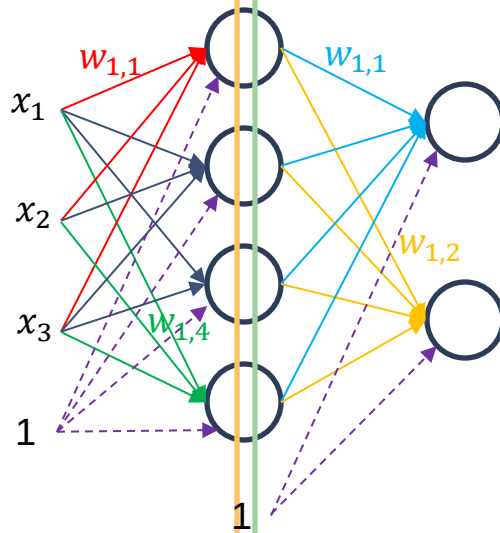
Multilayer Perceptron: Fundamental Concepts

- ❑ Each layer of neurons is fully connected to the subsequent layer.
- ❑ Each arrow represents a connection in the network.
- ❑ The connection within a network can be represented using matrices.

$$W_{ih} = \begin{bmatrix} w_{1,1} & \dots & w_{1,4} \\ \vdots & & \vdots \\ w_{3,1} & \dots & w_{3,4} \end{bmatrix}$$

Weights of
the 1st
neuron in the
hidden layer

Weights of
the 4th
neuron in the
hidden layer



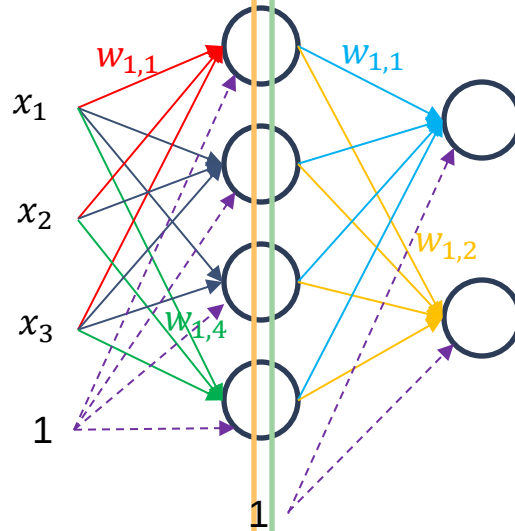
$$W_{ho} = \begin{bmatrix} w_{1,1} & w_{1,2} \\ \vdots & \vdots \\ w_{4,1} & w_{4,2} \end{bmatrix}$$



Multilayer Perceptron: Fundamental Concepts

- ❑ Each layer of neurons is fully connected to the subsequent layer.
- ❑ Each arrow represents a connection in the network.
- ❑ The connection within a network can be represented using matrices.

$$b_{ih} = [b_{1,1} \dots b_{1,4}]$$

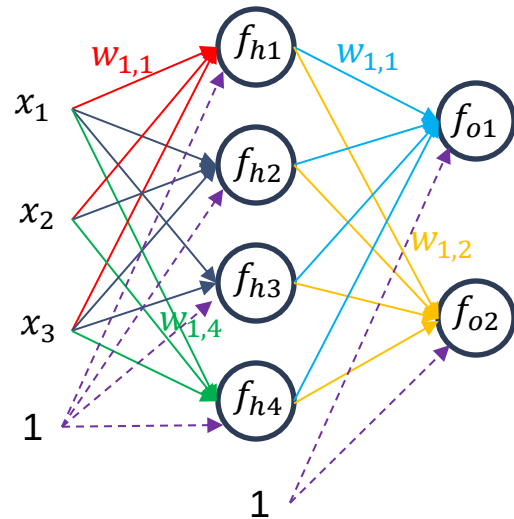


$$b_{ho} = [b_{1,1} \quad b_{1,2}]$$



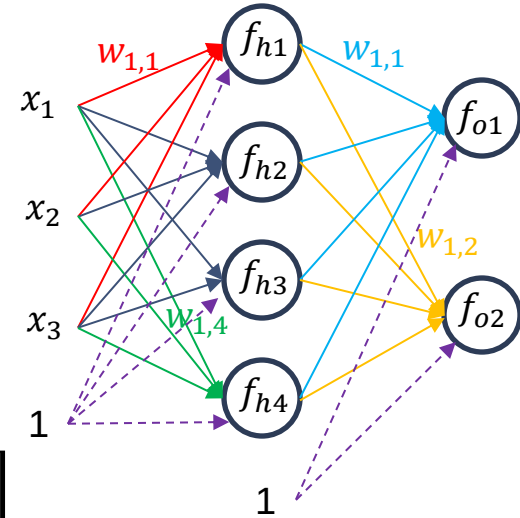
Multilayer Perceptron: Fundamental Concepts

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \times \begin{bmatrix} w_{1,1} & \dots & w_{1,4} \\ \vdots & & \vdots \\ w_{3,1} & \dots & w_{3,4} \end{bmatrix} + \begin{bmatrix} b_{1,1} & \dots & b_{1,4} \end{bmatrix} =$$
$$= \left[\sum_{k=1}^3 x_k w_{k,1} + b_{1,1}, \dots, \sum_{k=1}^3 x_k w_{k,4} + b_{1,4} \right]$$





Multilayer Perceptron: Fundamental Concepts

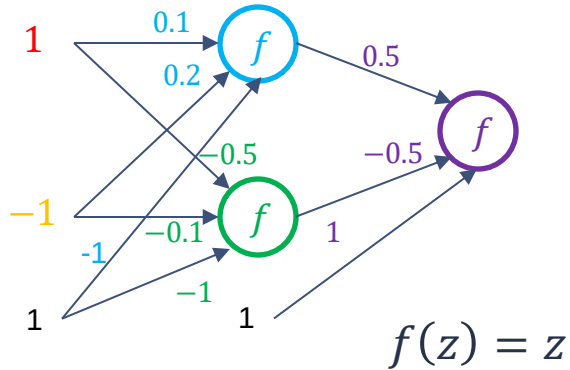


$$\left[f_{h1}(\sum_{k=1}^3 x_k w_{k,1} + b_{1,1}), \dots, f_{h4}(\sum_{k=1}^3 x_k w_{k,4} + b_{1,4}) \right] \times \begin{bmatrix} w_{1,1} & w_{1,2} \\ \vdots & \vdots \\ w_{4,1} & w_{4,2} \end{bmatrix} + \begin{bmatrix} b_{1,1} & b_{1,2} \end{bmatrix}$$



Multilayer Perceptron: Fundamental Concepts

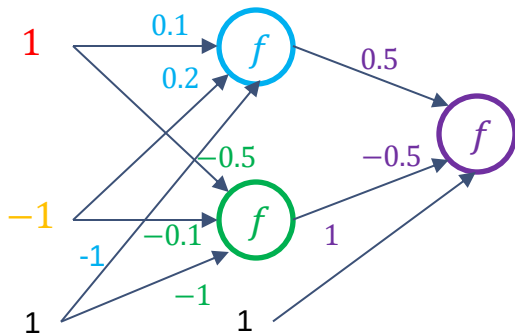
- Exercise: write in a matrix notation the calculation of the forward pass and calculate the output of the NN defined as follows:





Multilayer Perceptron: Fundamental Concepts

- ❑ Exercise: write in a matrix notation the calculation of the forward pass and calculate the output of the below NN:



$$x = [1, -1]$$

$$w_1 = \begin{bmatrix} 0.1 & -0.5 \\ 0.2 & -0.1 \end{bmatrix}$$

$$b_1 = [-1, -1]$$

$$w_2 = \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix}$$

$$b_2 = 1$$

$$output = f(f(xw_1 + b_1)w_2 + b_2) = (xw_1 + b_1)w_2 + b_2$$

$$xw_1 + b_1 = [1, -1] \begin{bmatrix} 0.1 & -0.5 \\ 0.2 & -0.1 \end{bmatrix} + [-1, -1] = [-1.1, -1.4]$$

$$(xw_1 + b_1)w_2 + b_2 = [-1.1, -1.4] \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix} + 1 = 1.15$$

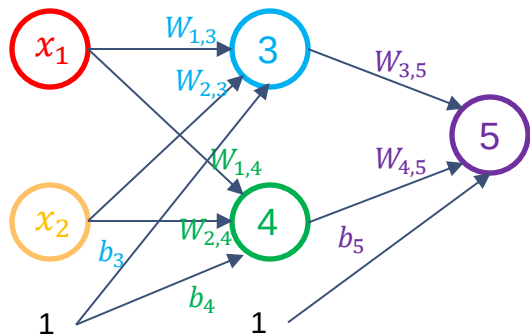


Multilayer Perceptron: Fundamental Concepts

- ❑ The goal of the neural network is to find a function that best maps a set on inputs to their correct outputs.
- ❑ For example, for a classification model, $y = f^*(x)$ maps an input x to a category y .
- ❑ The network defines a mapping $y = f(x; \theta)$ and learns the value of the parameters θ that results in the best function approximation.



Multilayer Perceptron: Feed-forward example



x_i - i th input to the NN

in_i - aggregated weighted input to the i th neuron

o_i - output of the i th neuron

f_i - activation function of the i th neuron

$$o_5 = f_5(\overbrace{o_3 W_{3,5} + o_4 W_{4,5} + b_5}^{in_5}) =$$
$$f_5(\underbrace{f_3(x_1 W_{1,3} + x_2 W_{2,3} + b_3)}_{in_3} W_{3,5} + \underbrace{f_4(x_1 W_{1,4} + x_2 W_{2,4} + b_4)}_{in_4} W_{4,5} + b_5)$$

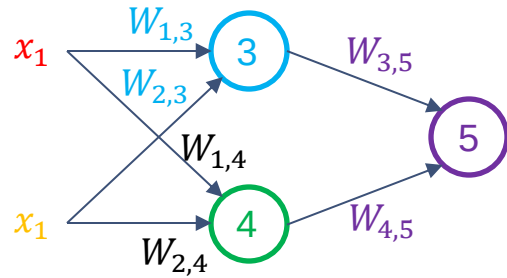


Multilayer Perceptron : Error calculation

- ☐ How well did the neural network do in estimating the output value for the provided input?
- ☐ Loss function $MSE = \frac{1}{n} \sum_{j=1}^n (y^j - \hat{y}^j)^2$ where y is the true value and $\hat{y}$ is the predicted value
- ☐ For a single input x , the error can be calculated as $(y - \hat{y})^2$
- ☐ Depending on the problem (i.e. trying to predict a single value – regression, or a categorical value - classification), we use different error/loss function.
- ☐ We will learn about different loss function later in the course.



Multilayer Perceptron: Error calculation



$$\begin{aligned} \text{Error} &= (\text{target}_{\text{output}} - \text{estimated}_{\text{output}})^2 \\ &= (\text{target}_{\text{output}} - o_5)^2 \end{aligned}$$

$$\begin{aligned} o_5 &= f_5(\overbrace{o_3 W_{3,5} + o_4 W_{4,5} + b_5}^{in_5}) = \\ &= f_5\left(\underbrace{f_3(x_1 W_{1,3} + x_2 W_{2,3} + b_3)}_{in_3} W_{3,5} + W_{4,5} \underbrace{f_4(x_1 W_{1,4} + x_2 W_{2,4} + b_4)}_{in_4}\right) \end{aligned}$$

Python Programming Example



Multilayer Perceptron.ipynb

Coming Next:

Training Neural Networks: Gradient
Descent and Backpropagation